



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: I

**DEPARTMENT
OF
COMPUTER SCIENCE AND ENGINEERING (DS)**

PRACTICE WORK BOOK

SUBJECT CODE: BE01R00061

SUBJECT NAME: ENGG. GRAPHICS & DESIGN

Subject Code:	BE01R00061	Subject Name	Credit:	2
Semester:	I	Engineering Graphics & Design	Hours:	60
Department:	Computer Science and Engineering (DS)			

CHAPTER: 1

INTRODUCTION TO ENGINEERING GRAPHICS

Drawing instruments and accessories, BIS – SP 46. Use of plain scales, Diagonal Scales and Representative Fraction

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Which (identify) type of line is used to indicate long break line (continuous thin with zig-zag line)? Which (identify) type of line is used to indicate hidden lines (dashed medium thick line)? What is Representative Fraction/Scale Factor? (Mar-2022) [NLJIET]	3	CO1	RM
2.	Differentiate (Compare) in between Unidirectional system and Aligned system of dimensioning. (Jun-2019) [NLJIET]	3	CO1	AN
3.	Draw the figure to explain the aligned and unidirectional system of dimensioning. (Nov-2020) [NLJIET]	3	CO1	AN
4.	Draw Parallel, Chain and Combined Dimensioning for following figure. (Jul-2024) [NLJIET] 	3	CO1	AN
5.	Show (Compare) by sketches the difference between (i) aligned dimensioning and (ii) unidirectional dimensioning (Jan-2024) [NLJIET]	3	CO1	RM
6.	Define the functions with symbols: 1) continuous thin, 2) Chain thin and thick at ends and 3) Chain thin (Aug-2023) [NLJIET]	3	CO1	RM
7.	Define representative fraction (R.F.) of scale. Also list types of scale according to value of R.F. (Mar-2023) [NLJIET]	3	CO1	RM
8.	Define Representative Fraction. With reference to Representative Fraction, classify the scales. (Jan-2024), (Jun-2025-New) [NLJIET]	3	CO1	RM
9.	Define Representative Fraction (RF). Find RF if 5 cm in the drawing is equal to 10 km. (Nov-2020). [NLJIET]	3	CO1	AN
10.	For 100 cm of a line compare size of drawing length on basis of full scale, reducing scale & enlarged scale. (Jan-2019) [NLJIET]	3	CO1	AN
11.	Give illustration and application of types of lines. (Jan-2025-New) [NLJIET]	3	CO1	RM
12.	Enlist types of lines. (Jun-2025-New) [NLJIET]	3	CO1	RM

13.	Give difference between two dimensioning methods with drawing. (Jan-2025-New) [NLJIET]	3	CO1	RM
14.	Explain reducing scale. Give two practical applications of it. (Jan-2025) [NLJIET]	3	CO1	RM
15.	Identify the following types of line and write their use in engineering drawing: (Jan-2024) [NLJIET] 1)  2)  3)  4) 	4	CO1	RM
16.	Write at least one use of following drawing instruments: (Jul-2024) [NLJIET] (1) Mini drafter (2) Protector (3) Set squares (4) Compass (5) Divider (6) Roll-N-Draw (7) T-square (8) French curve	4	CO1	RM
17.	Draw symbol of following and write use of it. (1) Cutting Plane line (2) Dimension line (3) Break line (4) Section Line (Jan-2025) [NLJIET]	4	CO1	RM
18.	Bisect one of the interior angle of regular pentagonal plane having sides of 30mm. (Jan-2025) [NLJIET]	4	CO1	RM
19.	Differentiate between aligned system and unidirectional system of dimensioning. (Jun-2025-New) [NLJIET]	4	CO1	RM

QUESTIONS FOR DRAWING

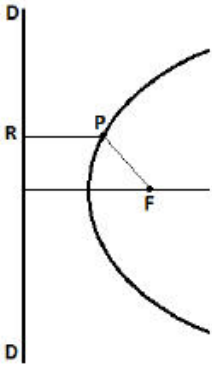
Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Draw regular square, pentagon and hexagon by UNIVERSAL METHOD of 35 mm side. (Jul-2024) [NLJIET]	3	CO1	RM
2.	Draw regular pentagon and hexagon of 30 mm side using UNIVERSAL METHOD (Jun-2025-NEW) [NLJIET]	3	CO1	RM
PLAIN SCALE				
3.	Explain Representative Fraction. Construct a plain scale of R.F. = 1:50 to show meters and decimetres and long enough to measure 8 meters. Indicate 5.7 m distance on it. (Aug-2022) [NLJIET]	3	CO1	UN
4.	Construct a scale of 1 cm = 1 m to read meters and decimetres and long enough to measure up to 14 meters. Show on this scale a distance equal to 12.4 meters. (Jan-2020) [NLJIET]	3	CO1	AP

5.	Construct a scale of 1:40 to read meters and decimetres and long enough to measure 6 m. Mark on it a distance of 4.7 m and 3.2 m.(Jan-2024) [NLJIET]	4	CO1	AP
6.	Construct a scale of 1:50 to read up to 6 metres in meters and decimetres. Show on it a distance of 4.9 metres. (Jan-2025-New) [NLJIET]	4	CO1	AP
7.	The distance between two towns (cities) is 250 km and is represented by a line of length 50mm on a map. Construct a scale to read 600 km and indicate a distance of 530 km on it. (Jun-2025-New) [NLJIET]	4	CO1	AP
8.	The length of the Khandala Tunnel on the Mumbai-Pune express way is 330 m, on the road map it is shown by 16.5 cm long line. Construct a plain scale to show meters and to measure up to 400 m. Show length of 290 m long on the express way. [NLJIET]	7	CO1	AP
DIAGONAL SCALE				
9.	Construct a diagonal scale of R.F. = 1/4000 to show 374 meters and long enough to measure up to 500 meters.(Aug-2022) [NLJIET]	3	CO1	AP
10.	Draw a diagonal scale of R.F. 1:5 showing decimetres, centimetres and millimetres and long enough to measure up to 8 decimetres. Show a distance of 5.35 dm. (Jun-2019)[NLJIET]	3	CO1	AP
11.	The distance between Ahmedabad and Surat is 200 km. In a railway map it is represented by a line 5 cm long. Find its R.F. Draw a diagonal scale to show single km. And maximum 600 km. Indicate on it following distances. (1) 222km (2) 569 km (Jul-2024)[NLJIET]	4	CO1	AN
12.	The distance between Delhi and Agra is 200 km. In a railway map it is represented by a line 5 cm long. Find its R.F. Draw a diagonal scale to show single km. and maximum 600 km. Indicate on it the distance of 322 km. (Aug-2022) [NLJIET]	4	CO1	AN
13.	Construct a scale to measure km, 1/8 km and 1/40 of a km, in which 1 km is represented by 4 cm. Mark on this scale a distance of 2.775 km. (Aug-2023) [NLJIET]	4	CO1	AN
14.	Construct a diagonal scale of RF=1/36 showing yard, foot and inch. Scale should be long enough to measure 5 yards. Indicate on it 3 yards, 2 feet and 9 inches.(Jan-2025-New) [NLJIET]	7	CO1	AP
15.	The distance between two trees on road is 876m. Show it on a scale of 1:6250. Also show distance of 653m distance between two vehicles parked on a side of the road.(Jan-2025) [NLJIET]	7	CO1	AP

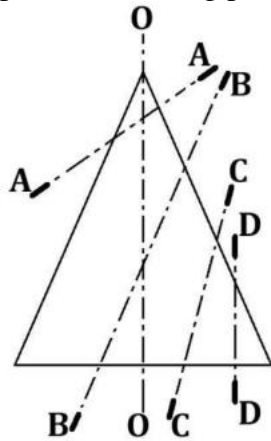
CHAPTER: 2**Engineering Curves**

Classification and application of Engineering Curves, Construction of Conics, Cycloidal Curves, Involute and Spirals

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Define Parabola, Ellipse and Hyperbola. (Jan-2024) [NLJIET]	3	CO1	RM
2.	Define eccentricity ratio. Write the values of eccentricity for (i) ellipse (ii) parabola and (iii) hyperbola (Aug-2022) [NLJIET]	3	CO1	RM
3.	In the figure shown here, show the conditions for having Ellipse, Parabola and Hyperbola for the following equation: PF/PR (Jan-2024) [NLJIET] 	3	CO1	RM
4.	List two applications (USE) of an Ellipse, Parabola and Hyperbola. (Jan-2019) [NLJIET]	3	CO1	AP
5.	Define the following terms: (1) cycloid (2) epicycloid (3) Hypocycloid [NLJIET]	3	CO1	RM
6.	Define the following curves: (i) Cycloid (ii) involute (Jun-2019) [NLJIET] (iii) Archimedean spiral	3	CO1	RM
7.	Give classification of Engineering Curves. (Jan-2025-New) [NLJIET]	3	CO1	RM
8.	Define ellipse. What is eccentricity? What is the value of eccentricity for ellipse? (Jan-2025) [NLJIET]	3	CO1	RM
9.	Define the following curves: involute, ellipse and cycloid. (Jun-2025-New) [NLJIET]	3	CO1	RM
10.	Define: (1) Involute (2) Spiral (3) Cycloid [NLJIET]	3	CO1	RM
11.	Enlist methods for drawing ellipse. (Jun-2025-New) [NLJIET]	3	CO1	RM
12.	Give Conic Terminology with sketch. (Jan-2025-New) [NLJIET]	4	CO1	RM
13.	Define the following terms: (1) conics (2) ellipse (3) parabola (4) hyperbola [NLJIET]	4	CO1	RM
14.	Give the definitions of the following curves: (i) Ellipse (ii) Involute (Nov-2020) [NLJIET]	4	CO1	RM
15.	Define the following terminology with respect to conics and also show them in a diagram: (Aug-2022) [NLJIET] (i) Focus (ii) Directrix (iii) Axis (iv) Vertex	4	CO1	RM

16. When we cut a cone using cutting planes as shown in figure, name the curves which are available as cut section of cone with particular cutting plane. (Jan-2025) [NLJIET]



4

CO1

RM

QUESTIONS FOR DRAWING

Sr. No.	QUESTIONS	Marks	CO	RBT Level
ELLIPSE				
1.	The foci of an ellipse are 110 mm apart. The minor axis is 70 mm long. Determine (Find) the length of major axis and draw half ellipse by rectangular method. (Mar-2022), (Aug-2022) [NLJIET]	4	CO1	AN
2.	Construct an ellipse when its major axis is 100 mm and minor axis 70 mm by using concentric circle method. (Aug-2022) [NLJIET] The foci of an ellipse are 110 mm apart. The minor axis is 70 mm long. Draw half ellipse by concentric circle method. (Mar-2022) [NLJIET]	4 4	CO1 CO1	AP AP
3.	Major axis of an ellipse is 120 mm and the distance between foci of ellipse is 80 mm. Draw an ellipse using oblong method. Draw tangent and normal to any point to the curve. (Jan-2024) [NLJIET]	4	CO1	AP
4.	The major and minor axes of an ellipse measure 100 mm and 70mm respectively. Draw an ellipse by the rectangle method. (Jan-2025-New) [NLJIET]	7	CO1	AP
5.	Draw an ellipse using concentric circle method. Given the distance between Foci is 90mm and length of major axis is 130mm. (Jan-2025) [NLJIET]	7	CO1	AP
6.	The foci of an ellipse are 120 mm apart. The minor axis is 72mm long. Determine (Find) the length of major axis and draw half ellipse by Oblong method [NLJIET]	7	CO1	AP
7.	Draw an ellipse having major axis 120mm and minor axis 80mm. Use Arc of Circle method. [NLJIET]	7	CO1	AP
8.	Construct an ellipse when the distance of the focus from the directrix is equal to 60mm and eccentricity is $\frac{2}{3}$. [NLJIET]	7	CO1	AP

PARABOLA

9.	A person travels in a curved path such that 25 mm distances from a fixed straight line and fixed point remain the same. Construct the traced curvature path by person and name the curve. (Aug-2023) [NLJIET]	4	CO1	AP
10.	A triangle ABC has sides AB = 75 mm, BC = 60 mm and CA = 75 mm. Draw a parabola passing through points A, B and C when side BC is horizontal. (Jun-2019) [NLJIET]	7	CO1	AP
11.	Draw a parabola by Rectangle method, given its base as 90 mm and the height of axis is 75 mm. [NLJIET]	7	CO1	AP
12.	Draw (Construct) a parabola by tangent method, given its base as 90 mm and the height of axis is 75 mm. (Jan-2020) [NLJIET]	7	CO1	AP
13.	Construct a parabola, when the distance of the focus from the directrix is 40 mm. Also draw a tangent and normal at a point on it 30 mm from F. (Aug-2022) [NLJIET]	7	CO1	AP

HYPERBOLA

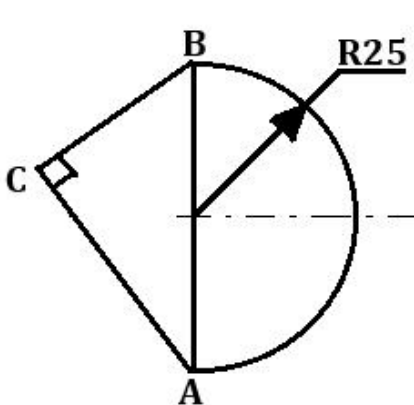
14.	A fixed point is 54 mm away from a fixed straight line. Draw (Construct) the locus of a point P moving in such a way that the ratio of its distance from the fixed straight line is 5:4. Name the curve. (Jan-2019) [NLJIET]	7	CO1	AP
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CYCLOID

15.	Draw (construct) a cycloid for a rolling circle, of 60 mm diameter rolling along a straight line without slipping. Take initial position of the tracing point at the bottom of the vertical centre line of the rolling circle. (Aug-2022) [NLJIET]	7	CO1	AP
16.	A wheel rolls over the horizontal straight road and covers 220 mm distance in one rotation. Draw the path traced by the point P which is initially at the point of contact between the wheel and the road. (Aug-2023) [NLJIET]	7	CO1	AP
17.	A thin circular disc of 50 mm diameter is allowed to roll without slipping from upper edge of sloping board which is inclined at 15° with the horizontal plane. Draw (Construct) the curve traced by the point on the circumference of the disc. Consider the point is at contact surface of disc and sloping board initially. (Jan-2024) [NLJIET]	7	CO1	AP

EPICYCLOID

18.	Draw an epicycloid with rolling circle diameter 50 mm and directing circle diameter 150 mm. Draw tangent and normal at a point on the curve 110 mm from the centre of the directing circle. (Jun-2019) [NLJIET]	7	CO1	AP
19.	Draw epicycloid of a 40 mm diameter circle, which rolls outside on another circle of 150 mm diameter for one revolution clockwise. Draw a tangent and normal to it at a point 95 mm from the centre of the directing circle. (Nov-2020) [NLJIET]	7	CO1	AP
20.	A circle of 50 mm diameter rolls on the circumference of another	7	CO1	AP

	circle of 150 mm diameter and outside it. Draw the locus of the point P on the circumference of the rolling circle for one complete revolution of it. Take initial position of point P at the contact point between two circles. Name the curve. (Aug-2022)[NLJIET]			
HYPOCYCLOID				
21.	Draw the hypocycloid taking rolling circle radius as 20mm and the directing circle radius as 100mm.[NLJIET]	7	CO1	AP
INVOLUTE				
22.	Construct the involute of a circle of 30mm diameter for one turn. [NLJIET]	4	CO1	AP
23.	Construct the involute of circle of 30 mm diameter for one turn. Draw tangent and normal to the involute at any point on it. (Mar-2022)[NLJIET]	7	CO1	AP
24.	Construct one complete turn of an involute of a circle having diameter 30 mm. Also draw tangent and normal to the curve S. The point S is a point on the involute and it is at a distance of 45mm from centre of the circle. (Mar-2023) [NLJIET]	7	CO1	AP
25.	Construct an Involute of a regular pentagon of 25 mm sides. (Jun-2017)[NLJIET]	7	CO1	AP
26.	Construct the Involute of hexagon of 25 mm sides. (Aug-2023) [NLJIET]	7	CO1	AP
27.	Construct the involute of a hexagon of side 20 mm. Draw the tangent and normal to the involute at any point.(Mar-2022) [NLJIET]	7	CO1	AP
28.	A pole is of a shape of half hexagon and semicircle. A string is to be wound having length equal to the pole perimeter Draw (Construct) path of free end P of string when wound completely.(Take hexagon 30 mm sides and semicircle of 60 mm diameter.)(Jul-2024) [NLJIET]	7	CO1	AP
29.	Draw involute of a string unwound for the given figure from point C, where AC = 40 mm and BC = 30 mm (Jan-2025) [NLJIET]	7	CO1	AP
				
ARCHEMEDIAN SPIRAL				
30.	A point P moves towards another point O, 90 mm from it, and reaches it during 1.5 revolutions around it in clockwise direction. Its movement towards O is uniform with its movement around it.	7	CO1	AP

	Draw (Construct) the curve traced out by the point P and name it.(Jan-2019) [NLJIET]			
31.	A point P is 120 mm away from the fixed point pole O. A point P moves towards pole O and reaches the position 'P' in one convolution where OP is 22 mm. The point P moves in such a way that its movement towards fixed point O, being uniform with its movement around fixed point pole O. Draw(Construct) the curve traced out by the point P. Name the curve. (Aug-2022) [NLJIET]	7	CO1	AP
32.	Draw an Archimedean spiral of 1½ convolutions, the greatest and least radii being 115 mm and 15 mm respectively. Draw a tangent and a normal to the spiral at a point 63 mm from the pole. (Aug-2023) [NLJIET]	7	CO1	AP
33.	Draw an Archimedean spiral of 1.5 convolutions, the largest & the smallest radius being 55 mm & 10 mm respectively. Draw tangent & normal to the spiral at a point 40 mm from the pole (Jun-2025-New) [NLJIET]	7	CO1	AP

CHAPTER: 3

PROJECTION OF POINTS AND LINES

Introduction to principal planes of projections, Projections of the points located in same quadrant and different quadrants, Projections of line with its inclination to one reference plane and with two reference planes. True length and inclination with the reference planes.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
QUESTIONS FOR DRAWING				
Sr. No.	QUESTIONS	Marks	CO	RBT Level
	TOPIC-1:Projection of Point			
1.	Draw the projections of the following points on the same X-Y line.(Mar-2022)[NLJIET] (1) Point A 25 mm below H.P. and 20 mm in front of V.P. (2) Point B 35 mm above H.P. and 40 mm in front of V.P. (3) Point C on V.P. and 30 mm above H.P.	3	CO2	UN
2.	Draw projection of following points. (i) Point R is 10 mm behind V.P. & 20 mm above H.P. (ii) Point S is in H.P. & 22 mm in front of V.P. (iii) Point T is 15 mm in front of V.P & 25 mm below H.P. (Jan-2019) [NLJIET]	3	CO2	UN
3.	Draw the projections of the following points on the same x-y line: (Jun-2019) [NLJIET]	3	CO2	UN

	(i) Point A is 20 mm above the HP and 20 mm behind the VP. (ii) Point B is 40 mm above HP and 10 in front of VP. (iii) Point C is 25 mm below HP and 40 mm behind VP.			
4.	Draw the projections of the following points on the same X-Y line. (a) Point 'A' is 20 mm below H.P and 20 mm in front of VP. (b) Point 'B' is 30 mm above H.P and 40 mm in front of VP. (c) Point 'C' is on VP and 30 mm above HP. (Nov-2020) [NLJIET]	3	CO2	UN
5.	Draw following projection of points: (Jul-2024) [NLJIET] (1) Point 'A' is 20 mm above HP and 25 mm in front of VP (2) Point 'B' is 35 mm behind VP and 30 mm above HP. (3) Point 'C' is in the HP and 15 mm In front of VP.	3	CO2	UN
6.	Draw the projection of following points on the same X-Y line. (Mar-2023) [NLJIET] (1) Point A is 20 mm above HP and 30 mm behind V.P. (2) Point B is on HP and 40 mm in front of V.P. (3) Point C is 10 mm below HP and 25 mm behind V.P.	3	CO2	UN
7.	Draw the projections of the following points on the same ground lines, keeping the projectors 15 mm apart: (Aug-2023) [NLJIET] (1) A in the H.P. and 20 mm behind V.P. (2) B 25 mm below the H.P. and 25 mm behind V.P. (3) D 40 mm below H.P. and 25 mm in front of V.P.	3	CO2	UN
8.	Draw following projection of points: (Jan-2025-New) [NLJIET] (1) Point 'A' is 20 mm above HP and 25 mm In front of VP. (2) Point 'B' is 35 mm behind VP and 30 mm above HP. (3) Point 'C' is in the HP and 15 mm in front of VP	3	CO2	UN
9.	Draw the projection of following points. (Jan-2025) [NLJIET] 1. Point A is 20 mm above HP and 35 mm behind VP 2. Point B is 20 mm below HP and 35 mm in front of VP 3. Point C is in VP and 15 mm above HP	3	CO2	UN
10.	Draw following projection of points: (Jun-2025-New) [NLJIET] (1) Point 'A' is 25 mm above HP and 20 mm In front of VP. (2) Point 'B' is 35 mm behind VP and 40 mm above HP. (3) Point 'C' is in the HP and 10 mm In front of VP	3	CO2	UN
11.	Draw projection of following points: (Jan-2024) [NLJIET] 1. Point A is in HP and in VP 2. Point B is in VP and 25 mm below HP 3. Point C is 10 mm below HP and 20 mm in front of VP 4. Point D is 20 mm above HP and 20 mm in front of VP	4	CO2	UN
12.	Draw the projection of following points on the same X-Y line. [NLJIET] (1) Point A is 25 mm above HP and 30 mm behind V.P. (2) Point B is 20 mm above HP and on V.P. (3) Point C is 10 mm below HP and 25 mm behind V.P.	4	CO2	UN

	(4) Point D is on HP and 40 mm in front of V.P.			
TOPIC-2: Projection of Line				
	Group-I (inclined/perpendicular/parallel to reference plane)			
1.	Draw (Construct) Projections of the following lines. (i) Line MN 50 mm is in 1 st quadrant and parallels both HP and VP. (ii) Line PQ 35 mm is in 3 rd quadrant and remains perpendicular to V.P. and parallel to H.P. (Jan-2019) [NLJIET]	3	CO2	UN
2.	A line PQ 65 mm long is parallel to HP and 40mm below HP Its two ends P and Q are 25mm and 45mm behind VP respectively. Draw the Projections and find inclination of line with VP.[NLJIET]	3	CO2	UN
3.	Draw projection of line CD is 50 mm long, when it is not contained by any plane and parallel to both planes. A line CD is 20 mm above H.P. and 25 mm in front of V.P. (Mar-2022) [NLJIET]	3	CO2	UN
4.	A line PQ 70 mm long is parallel to VP and 30° inclined to HP. The end P is 30 mm above HP and 20 mm in front of VP. Draw the Projections.(Aug-2022) [NLJIET]	3	CO2	UN
5.	A line PQ 130 mm long is parallel to the V.P. and inclined at 60° to H.P. The end nearest to the H.P. is 30 mm from it and 40 mm from the V.P. Draw the projection of line PQ. (Mar-2022) [NLJIET]	3	CO2	UN
6.	A line MN, 60 mm long, is in V.P. It makes an angle of 30° with the H.P. The end point M is 15 mm above H.P. Draw projections of line MN. Also measure Plan length of the line. (Mar-2023) [NLJIET]	4	CO2	UN
7.	A line AB, 75 mm long, is parallel to VP and inclined to the HP, by an angle 45°. Point A is 30 mm below HP and 20 mm in front of VP. Point B is in the first quadrant. Draw the projections of the straight line AB.(Jan-2019) [NLJIET]	4	CO2	UN
Group-II: (Inclined to both reference plane)				
8.	A line CD, 80 mm long is inclined at 45° to H.P. and 30° to the V.P., its end C is in H.P. and 40 mm in front of V.P. Draw (Construct) the projections. (Aug-2023) [NLJIET]	3	CO2	UN
9.	A line PQ 100 mm long is inclined at an angle of 45° to H.P. and 30° to V.P. One of its end point 'P' is in H.P. as well as VP. Determine (Find) its apparent inclination with V.P. (Aug-2022) [NLJIET]	4	CO2	AN
10.	A line AB 75mm long is incline at an angle of 45degree to H.P. and 30 degree to V.P. One of its end point A is in H.P. as well as in V.P. Determine its apparent angle with V. P. [NLJIET]	4	CO2	AN

11.	A straight line AB 110 mm long is inclined at 30° to H.P. and 45° to V.P. One of its end A is 40 mm from the V.P. and 20 mm above H.P. Draw the projection of line AB. (Mar-2022) [NLJIET]	7	CO2	UN
12.	A line AB is 100 mm long. It is inclined at 40° to the HP and 30° to the VP. The end A is 10 mm above HP and 25 mm in front of VP. Assuming the end B in the first quadrant, draw the projections of the line AB. (Jun-2019) [NLJIET]	7	CO2	UN
13.	The end P of a line PQ 120 mm long is 30 mm above HP and 60 mm behind VP. The line is inclined at angle of 30° with the reference planes of the projection (inclined at 30° to both HP & VP) The point Q is below HP and behind VP. Draw the projections of line PQ and locate the point Q. (Jan-2020) [NLJIET]	7	CO2	AN
14.	A line AB, 80mm long is inclined to HP at 30° and to VP at 45° . The end A is in HP and VP. Draw projection of line AB. Find length of top view and front view of line and inclination of them with HP and VP.(Jan-2024)[NLJIET]	7	CO2	AN
15.	A line AB, 75 mm long is inclined at an angle 35° to the H.P. and 55° to the V.P. Its end point 'A' is on the H.P. and 15 mm in front of the V.P. Draw the projections of the line AB assuming it to be in the first quadrant. (Aug-2022) [NLJIET]	7	CO2	UN
16.	A line AB 100 mm long is inclined to HP at 45° and inclined to VP at 30° . Draw front and top views of line and determine their lengths. Also determine the perpendicular distance of end B from both HP and VP. (Nov-2020) [NLJIET]	7	CO2	UN
17.	A Line AB, 90 mm long, is inclined to H.P. by 30° and inclined to V.P. by 45° . The line is in first quadrant with Point A 15 mm above H.P. and 25 mm in front of V.P. Draw the projection of line AB.[NLJIET]	7	CO2	UN
Group-III (Location of both ends)				
18.	A line EF, 75mm long has its end E 20mm below HP and 25mm behind VP. The end F is 50mm below HP and 65mm behind VP. Draw the projections of line EF and find its inclinations with HP and VP.[NLJIET]	7	CO2	UN
19.	A line AB, 75mm long has its end A 15mm above HP and 10mm in front of VP. The end B is 45mm above HP and 50mm in front of VP. Draw the projections of line AB and find its inclinations with HP and VP.[NLJIET]	7	CO2	UN
Group-IV: (EPD)				
20.	The distance between the end projectors of a straight line AB is 60 mm. Point A is 5 mm above HP and 30 mm in front of VP. Point B is 40 mm above HP and 50 mm behind VP. Draw (Construct) the projections and find the inclination of straight line AB with HP and VP and the true length of the line. (Jun-2019) [NLJIET]	4	CO2	UN
21.	Point A is 20 mm above HP and 30 mm in front of VP and point B is in the HP and 40 mm behind the VP. The distance between	4	CO2	UN

	in their projectors is 50 mm. Draw the projections of the points. Also draw straight lines joining their top and front views. (Jan-2020) [NLJIET]			
22.	Two lemons on a tree, planted near the compound wall of a bungalow are 1.0 m and 1.25 m above the ground and 0.5 m and 0.75 m from a 15 cm thick compound wall but on opposite sides of it. The distance between lemons measured along the ground and parallel to the wall is 1.0 m. Determine (Find) the real distance between the centers of two lemons. (Jun-2019) [NLJIET]	4	CO2	AN
23.	Two lemons on a tree, planted near the compound wall of a bungalow are 1.0 m and 1.25 m above the ground and 0.5 m and 0.75 m from a 15 cm thick compound wall but on opposite sides of it. The distance between lemons measured along the ground and parallel to the wall is 1.0 m. Determine (Find) the real distance between the centers of two lemons. (Jan-2025-New) [NLJIET]	7	CO2	AN
24.	The distance between the end projectors of a straight line PQ is 60 mm. The line makes 30° and 45° angles with HP and VP, respectively. The end P is 30 mm below HP and 50 mm in front of the VP. Draw its projections when end Q is in third quadrant. Find TL of the line. (Jan-2019) [NLJIET]	7	CO2	UN
25.	The line AB has its end A, 15 mm above H.P. and 10 mm in front of V.P. The end B is 60 mm above H.P. The distance between the end projectors is 50 mm. The plan of the line is inclined at 25° to X-Y line. Draw the projections of line. Find true length and inclination of the line with H.P. (Mar-2023) [NLJIET]	7	CO2	AN
26.	The distance between the end projectors of a straight-line PQ is 130mm. The end P is 40mm below HP and 25mm in front of the VP. Q is 75 mm above HP and 30 mm behind VP. Draw its projections. Find TL of the line. (Jun-2025-New) [NLJIET]	7	CO2	AN
Group-V (PL/EL)				
27.	Point P of a straight line PQ is 25 mm above H.P. and point Q is 65 mm in-front of V.P. The line makes an angle of 30° with H.P. and its plan is at 45° to the XY line. Draw the projections of the line if the plan length is 70 mm. Also find the true length of the line and the angle made by the line with V.P. (Jan-2019) [NLJIET]	7	CO2	AN
28.	Plan and elevation of a line AB, 80 mm long, measure 60 mm and 72 mm respectively. End A of the line is in HP and end B is in VP. Draw its projections, assuming the line to lie in first quadrant. (Jan-2020) [NLJIET]	7	CO2	UN
29.	Line AB 75 mm long makes 45° inclination with VP while its FV makes 55° . End A is 10 mm above HP and 15 mm in front of VP. If line is in 1 st quadrant draw its projections and find its inclination with HP. (Jul-2024) [NLJIET]	7	CO2	UN

30.	The top view of a 75 mm long line AB measures 65 mm, while its front view measures 50 mm. Its one end 'A' is in HP and 12 mm in front of VP. Draw the projections of AB and determine its inclination with HP and VP. (Nov-2020) [NLJIET]	7	CO2	UN
31.	A line PQ 85 mm long has its end 'P' 10 mm above HP and 15 mm in front of VP. The top view and front view of line PQ are 75 mm and 80 mm respectively. Draw its projections. Also determine the true and apparent inclinations of the line. (Nov-2020) [NLJIET]	7	CO2	UN
32.	A room is of size 6.5m L, 5m D, 3.5m high. An electric bulb hangs 1m below the centre of ceiling. A switch is placed in one of the corners of the room, 1.5m above the flooring. Draw the projections and determine real (Find out) distance between the bulb and switch. (Jul-2024) [NLJIET]	7	CO2	AN
33.	The top view of a 75mm long line AB measures 65mm, while its front view measures 50mm. Its one end A is in HP and 12mm in front of VP. Draw the projections of AB and determine its inclination with HP and VP. (Jan-2025) [NLJIET]	7	CO2	AN
34.	The elevation of line AB, 80 mm long, measures 55 mm. The end A is 20 mm above H.P. and 10 mm in front of V.P. Draw the projection of the line and find its true inclination with H.P. and V.P, if the end B is 25 mm below H.P. and is behind V.P (Jun-2025-New) [NLJIET]	7	CO2	AN
Group-VI (Three-point)				
35.	A 100 mm long line PQ is inclined at 30° to H.P. and 45° to the V.P. Its mid-point is 35 mm above the H.P. and 50 mm in front of V.P. Draw its projections. (Aug-2023) [NLJIET]	3	CO2	UN
36.	A line PQ, 100 mm long, is inclined at 30° to the HP and 45° to the VP. Its mid-point M is in the VP and 20 mm above the HP. Draw its projections, when its end P is in the first quadrant and Q is in the third quadrant. (Jan-2019) [NLJIET]	7	CO2	UN
37.	A line AB has a point P on it such that AP: PB = 1: 2. The end A is in the first quadrant and it is 20 mm above H.P. while the end B is in the V.P. The point P is 35 mm from the H.P. The line is inclined at 30° to the H.P. and the elevation length of the line is 70 mm. Draw the projections of the line AB and the point P. Find the true length, the plan length and the inclination of the line with V.P. (Jan-2025-New) [NLJIET]	7	CO2	UN
Group-VII (theta+ phi=90°)				
38.	Draw projections of line AB=80 mm inclined to 60° HP & 30° VP and parallel to PP. (Jul-2024) [NLJIET]	4	CO2	UN
39.	A line PQ 70 mm long is parallel to profile plane and inclined at 30° to HP. Point P is 10mm Above HP and point Q is in VP. Draw its projections. (Jan-2024) [NLJIET]	4	CO2	UN
40.	A line PQ 70 mm long has its end P in VP and end 'Q' in HP. Line is inclined to HP by 60° and VP by 30°. Draw the projections. (Jan-2025-New) [NLJIET]	4	CO2	UN

41.	Draw projections of line AB=85 mm inclined to 60° HP & 30° VP.(Jun-2025-New) [NLJIET]	4	CO2	UN
42.	Draw the projections of a straight line 80 mm long inclined at 60° to H.P and 30° to V.P. with end A in the H.P. and the end B in the V.P. (Mar-2023) [NLJIET]	7	CO2	UN

CHAPTER: 4**PROJECTION OF PLANES**

Projection of planes (polygons, circle and ellipse) with its inclination to one reference plane and with two reference planes.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
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QUESTIONS FOR DRAWING

Sr. No.	QUESTIONS	Marks	CO	RBT Level
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Group-I: Square Plane

1.	A square lamina of 40 mm side rests on one of its sides on HP. The lamina makes 30° with HP and the side on which it rests makes 45° with VP. Draw its projections. (Nov-2020) [NLJIET]	3	CO2	AP
2.	A square plate PQRS, edge 25 mm, is in space with one of its corners in VP. Surface of the plate makes 50° with VP and it is perpendicular to HP. Draw its projections. (Jan-2019) [NLJIET]	4	CO2	AP
3.	A square plate, 40 mm side, is resting on the H.P. on one of its sides. The plate is inclined to H.P. such a way that its top view looks like a rectangle with 20 mm shorter side. Draw its projections. (Mar-2023) [NLJIET]	4	CO2	AP
4.	A square ABCD of 50 mm side is resting on HP on one of its side such that it appears as a rectangle of 35 mm X 50 mm. Draw its projection and find its inclination with HP. (Jan-2024) [NLJIET]	4	CO2	AP
5.	A square plane ABCD is of 30 mm side. It is kept on V.P. on one of its corner and it is inclined to V.P. at an angle of 30°. The surface of the plane is perpendicular to H.P. Draw the projection of plane (Mar-2022) [NLJIET]	7	CO2	AP

Group-II: Rectangular Plane

6.	A rectangular plane ABCD having 60 mm X 30 mm size is parallel to V.P., and perpendicular to H.P., and P.P. Draw (Construct) the projections of the rectangle when it is 40 mm in	3	CO2	AP
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	front of V.P., and one of the smaller sides is parallel to H.P. and 20 mm above it. (Aug-2022) [NLJIET]			
7.	A rectangle ABCD 60 mm x 40 mm, is parallel to HP with one of its sides inclined at 30° to VP and the end of the side near to VP is 15 mm in front of the VP and 30 mm above HP. Draw (Construct) its projections. (Jan-2020) (Jul-2024) [NLJIET]	4	CO2	AP
8.	A rectangular plane ABCD having 60 mm x 30 mm size is parallel to H.P. and perpendicular to V.P. and P.P. Draw (Construct) the projections of the rectangular when it is 40 mm above H.P. and one of the longer sides is parallel to V.P. and 20 mm in front of it. (Mar-2022) [NLJIET]	4	CO2	AP
9.	A rectangular plate PQRS 25 mm x 40 mm size, is in space with shorter edge parallel to H.P. and 15 mm above it. Plate PQRS is perpendicular to V.P. and inclined to H.P. by such an angle so that its plan becomes square. Draw the projections and find out its inclination with H.P. (Jan-2025-New) [NLJIET]	4	CO2	AP
10.	A thin rectangular plate of side 40mm x 20mm has its shorter side in HP. Project its front view when its top view is a perfect square of 20mm side. (Jan-2025) [NLJIET]	4	CO2	AP
Group-III: Rhombus Plane				
11.	ABCD is a rhombus of diagonals AC 110 mm and BD 70 mm. Its corner A is in the H.P. and the plane is inclined to the H.P. such that the plan appears to be a square. Draw the projections of the plane and find its inclination with H.P. (Jan-2025-New) [NLJIET]	4	CO2	AP
12.	A rhombus of 30mm and 50mm diagonal is resting on HP on one of its corner such that it appears as square of 30mm diagonals. Find its inclination with HP. (Jan-2025) [NLJIET]	4	CO2	AP
13.	Draw the projections of rhombus diagonals 96 mm and 48 mm long. The smaller diagonal is parallel to both the reference planes, while the other diagonal is inclined at 30° to the HP and has one of its end points in the HP. Keep the center of rhombus 56 mm in front of the VP. (Jan-2020) [NLJIET]	7	CO2	AP
14.	Draw the projections of a rhombus having 100 mm and 40 mm long diagonals. The bigger diagonal is inclined at 30° to H.P. with one of the end point in H.P. and the smaller diagonal is parallel to both the planes. (Aug-2023) [NLJIET]	7	CO2	AP
Group-IV: Pentagonal Plane				
15.	A pentagonal plate, side 25 mm, is inclined to H.P. by 30°. It is perpendicular to V.P. and resting on HP on its one of the side. Draw projections of the plate. (Mar-2023) [NLJIET]	3	CO2	AP
16.	A regular pentagon laminate of 30 mm each side is resting on HP on one of its sides with its surface making 45° with HP. Draw (Construct) its projection when the side in HP makes an angle 30° with VP. (Nov-2020) [NLJIET]	4	CO2	AP
17.	ABCDE is a regular pentagonal plate of 40 mm sides, has its corner A on the HP. The plate is inclined to the HP such that the	7	CO2	AP

	plan length of the edges AB and AE is each 35 mm. The side CD is parallel to both the reference planes. Draw the projections of the plate and find its inclination with the HP. (Jun-2019) [NLJIET]			
18.	A regular pentagon of 30 mm sides is resting on HP on one of its sides with its surface 45° inclined to HP. Draw its projections when the side in HP makes 30° angles with VP. (Jul-2024) [NLJIET]	7	CO2	AP
19.	A pentagonal plane having edges 25 mm is placed on one of its corners on HP such that the surface makes an angle 30° with HP and side opposite to that corner on which the plane rests inclined at 60° to VP. Draw the top and front views of the plane. (Jan-2025) [NLJIET]	7	CO2	AP
20.	A pentagonal plate, side 25 mm is resting on H.P. on one of its corners with opposite edge to the corner making 30° with V.P. The plate is inclined to H.P. by 45° . Draw the projections of regular pentagonal plate. (Jun-2025-New) [NLJIET]	7	CO2	AP
Group-V: Hexagonal Plane				
21.	A hexagonal plate, side 25 mm, is inclined to H.P. by 45° . It is resting on HP on one of the corner. The plate is perpendicular to V.P. Draw (Construct) projections of the plate. (Mar-2023) [NLJIET]	3	CO2	AP
22.	The top view of a lamina whose surface is perpendicular to V.P. and inclined at an angle of 45° to H.P. appears as a regular hexagon of 30 mm side, having a side parallel to the reference line. Draw (Construct) the projections of plane and obtain its true shape. (Aug-2023) [NLJIET]	4	CO2	AP
23.	A hexagonal plane of 30 mm side, rests on the V.P. on an edge such that the surface is inclined at 45° to the V.P. Draw (Construct) its projections. (Aug-2023), (Jun-2025-New) [NLJIET]	4	CO2	AP
24.	A regular hexagon lamina ABCDEF has a side AB in VP and its side DE 45 mm in front of the VP and inclined to HP at 30° . Draw (Construct) its projections. Consider size of lamina as 50 mm. (Jan-2020) [NLJIET]	7	CO2	AP
Group-VI: Triangular Plane				
25.	A 30° - 60° set square has its shortest side 50 mm long and is in the H.P. The Top view of the set square is isosceles triangle. Draw the projections of the set square find its inclination with the H.P. (Jan-2025-New) [NLJIET]	3	CO2	AP
26.	ABC is a triangle of sides AB = 75 mm, BC = 60 mm and CA = 45 mm. Its longest side is in H.P. Its surface makes an angle of 45° with the H.P. Draw its projections. (Jun-2025-New) [NLJIET]	4	CO2	AP
27.	An isosceles triangular XYZ having its base XY=50 and altitude 60mm is resting on H.P. on its base XY with its surface making angle of 45° degree to H.P. Draw projection of	4	CO2	AP

	plane.[NLJIET]			
28.	An isosceles triangular plate of 50 mm base and 75 mm altitude appears as an equilateral triangle of 50 mm in top view. Draw the projections of a plate if its 50 mm long edge is on the HP and inclined at 45° to the VP. What is the inclination of the plate with the HP? (Jun-2019) [NLJIET]	7	CO2	AN
29.	An isosceles triangular plate ABC has its base 45 mm and altitude 60 mm. It is so placed that the front view is seen as an equilateral triangle of 45 mm side and (i) base is inclined at 45° to HP, (ii) side is inclined at 45° to HP. Draw its plan when its corner A is on HP. (Jan-2019) [NLJIET]	7	CO2	AP
30.	A $30^\circ - 60^\circ$ set square of 60 mm longest side is so kept such that the longest side is in HP making an angle of 30° with VP. The set square itself is inclined at 45° to HP. Draw the projections of the set square. (Jan-2024) [NLJIET]	7	CO2	AP
31.	An isosceles triangle ABC having its base AB = 40 mm and altitude 60 mm is resting on the H.P. on its base AB. Draw the projections of the plane when its surface is inclined to H.P. at an angle of 45° and the base AB which is on the H.P. is making an angle of 50° to the V.P (Jan-2025-New) [NLJIET]	7	CO2	AP
Group-VII: Circular Plane				
32.	A circular plate, 50 mm diameter is resting on HP on one of the points of its periphery with surface of the plate perpendicular to VP and inclined to HP by 30° . Draw two projections of the circular plate. (Jun-2019) [NLJIET]	3	CO2	AP
33.	A circular plane of 50 mm diameter is perpendicular to HP and VP. The centre of plane is 30mm above HP and 35mm in-front of VP. Draw its projection. (Jan-2025) [NLJIET]	3	CO2	AP
34.	A circular plane having 50 mm diameter is resting with point A of its periphery on H.P. The surface of plane is inclined with 30° H.P. Draw the projection of this circular plane. (Mar-2022) [NLJIET]	4	CO2	AP
35.	A circular plate of diameter 50 mm is resting on HP on a point on the circumference with its surface inclined at 45° to HP and perpendicular to VP. Draw its projections. [NLJIET]	4	CO2	AP
36.	A circular disk, 50 mm in diameter, is resting on the H.P. on one of the point A of the circumference. Plane is inclined to the H.P. such a way that the top view of it is an ellipse of minor axis 40 mm. Draw its projections. (Mar-2023) [NLJIET]	4	CO2	AP
37.	A circular plane of 80 mm diameter has one of the ends of the diameter in the H.P. while the other end is in the V.P. The plane is inclined at 30° to the H.P. and 60° to the V.P. Draw its projections. (Aug-2023) [NLJIET]	7	CO2	AP
38.	A circular plane having the diameter 50 mm is resting with point A of its periphery on H.P. The surface of the plane is inclined to H.P. such that the plan of the plane becomes an ellipse with minor axis 30mm. Draw the projection of the plane when the plan of the diameter through point A is inclined at 30° to V.P. and the	7	CO2	AP

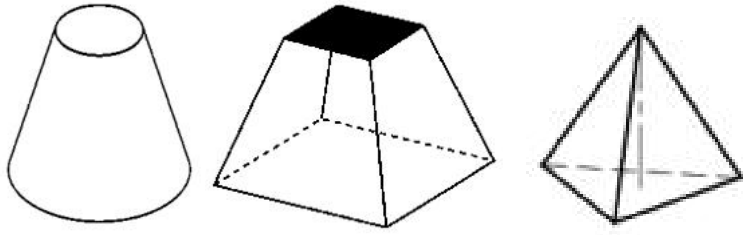
	centre of the plane is 40mm from V.P. Find the inclination of the plane with H.P. (Jun-2025-New) [NLJIET]			
Group-VIII: Semi Circular Plane				
39.	A semi-circular thin plate, of 60 mm diameter, rests on the H. P. on its diameter, the surface is inclined at 30° to the H. P. Draw the projections of the plate. (Jan-2025-New) [NLJIET]	4	CO2	AP
40.	A semi-circular plate has the 60 mm diameter straight edge in HP and inclined at 45° to VP. The surface of plate makes an angle of 30° to HP. Draw its projection. (Jan-2024) [NLJIET]	7	CO2	AP
41.	A semi-circular plate of 80 mm diameter has its straight edge in the VP and inclined at 45° to the HP. The surface of the plate makes an angle of 30° with the VP. Draw its projections. (Jan-2019) [NLJIET]	7	CO2	AP

CHAPTER: 5**PROJECTION OF SOLIDS, SECTION OF SOLIDS**

Classification of solids. Projections of solids (Cylinder, Cone, Pyramid and Prism) along with frustum with its inclination to one reference plane and with two reference planes, Section of such solids and the true shape of the section, Development of surfaces

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Define right solid, oblique solid and regular solid. (Jan-2019) [NLJIET]	3	CO3	RM
2.	Differentiate (Compare) between (i) Prism and pyramid (ii) Square pyramid and tetrahedron (iii) Cube and square prism (Jun-2019) [NLJIET]	3	CO3	UN
3.	Differentiate (Compare) between prism and pyramid. (Jan-2024) [NLJIET]	3	CO3	UN
4.	Give complete classification of solids. (Nov-2020) [NLJIET]	3	CO3	UN
5.	Classify solids with shapes of drawings. (Aug-2023), (Jun-2025-New) [NLJIET]	3	CO3	UN
6.	Draw the following sketches: Truncated Cylinder, Frustum of a cone and Frustum of a square pyramid. (Mar-2022), (Jan-2025-New) [NLJIET]	3	CO3	AN
7.	Identify following solids: (Jul-2024-NEW) [NLJIET] (i) (ii) (iii)	3	CO3	RM

				
8.	Define apparent shape and true shape with diagram. (Jan-2019) [NLJIET]	3	CO3	UN
9.	The Top View of a right cylinder resting on HP on its base is _____. (Jul-2024) [NLJIET]	1	CO3	UN
10.	Give different types of section planes. (Jan-2025-New) [NLJIET]	3	CO3	RM
11.	Define solid of revolution? Give three example of solid of revolution. (Jan-2025) [NLJIET]	3	CO3	RM
12.	Differentiate between prism and pyramid. (Jan-2025) [NLJIET]	3	CO3	RM
13.	Differentiate (Compare) between (Jun-2025-New) [NLJIET] (1) Square pyramid and tetrahedron (2) Cube and square prism	3	CO3	RM
14.	Differentiate between prism and pyramid. (Jun-2025-New) [NLJIET]	4	CO3	RM

QUESTIONS FOR DRAWING

Sr. No.	QUESTIONS	Marks	CO	RBT Level
	TOPIC-1: PROJECTION OF SOLIDS			
Group-I :PRISM				
1.	A square prism edge of base 35 mm and axis length 50 mm is resting on its base on the H.P. with an edge of base parallel to V.P. Draw the projection of prism. (Mar-2022) [NLJIET]	4	CO3	AN
2.	A square prism, side of base 30 mm and height 45 mm, is resting on H.P. on one of the corners of the base. The longer edge containing that corner on which it rests, is inclined at 45° to H.P. and top view of it is inclined at 30° to V.P. Draw the projections when top end of the prism is nearer to V.P. (Mar-2023) [NLJIET]	7	CO3	AN
3.	A hexagonal prism with side of base 30 mm and axis length 60 mm is resting on one of its base edge on HP such that its axis is inclined at 45° with HP and the side on which it is resting is inclined at 30° with VP. Draw the projections. (Nov-2020) [NLJIET]	7	CO3	AN
Group-II: PYRAMID				
4.	A square pyramid, side of base 40 mm and axis 60 mm is resting on its base on H.P. Draw its projections when (a) a side of the	4	CO3	AN

	base is parallel to V.P. (b) a side of the base is inclined at 30° to V.P. and (c) all the sides of the base are equally inclined to V.P. (Aug-2023) [NLJIET]			
5.	A right regular pentagonal pyramid, edges of base 25 mm and height 50 mm, has its base parallel to VP with one of its base edges in HP. Draw its projections. (FV and TV) (Jan-2020) [NLJIET]	4	CO3	AN
6.	A square pyramid, side of base 35 mm and axis length 50 mm is lying on the H.P. on one of its triangular faces. Draw the projections of the pyramid when the base edge contained by the triangular face on H.P. makes an angle of 45° on the V.P. keeping apex of the pyramid towards the observer. (Aug-2022), (Jun-2025-New) [NLJIET]	7	CO3	AN
7.	A square pyramid, side of base 30 mm and height 50 mm, is resting on HP with a side of its base. Draw the projections if the axis is inclined to 45° to HP and parallel to VP. (Jul-2024) [NLJIET]	7	CO3	AN
8.	A square pyramid having 30mm edge of the base and axis of 70mm long and axis parallel to VP and inclined at 60° to HP. Draw projection of pyramid if one of the edge of its base inclined at 30° to VP and apex is on HP and 40 mm away from VP. (Jan-2025) [NLJIET]	7	CO3	AN
9.	Draw the top view and front view of a rectangular pyramid of sides of base 40 mm x 50 mm and height 70mm when it lies on one of its larger triangular faces on HP. The longer edge of the base of the triangular face lying on HP is inclined at 60° to VP in the top view with the apex of the pyramid being nearer to VP. (Jan-2025) [NLJIET]	7	CO3	AN
Group-III: CONE				
10.	Draw the projections of a cone, base 60 mm diameter and axis 90 mm long when it is resting on the HP on a point on its base circle with the axis making an angle of 30° with the HP and parallel to the VP. (Jun-2019)[NLJIET]	3	CO3	AN
11.	A cone, diameter of base 55 mm and height 60 mm, is resting on HP on one of its generators with axis parallel to VP. Draw (Construct) the projections of cone. (Jan-2019) [NLJIET]	4	CO3	AN
12.	A cone 40 mm diameter and 50 mm axis is resting on one generator on HP. Draw its projections. (Jun-2025-New) [NLJIET]	4	CO3	AN
13.	A cone, diameter of base 60 mm and height 70 mm, has one of its generators in H.P. and making an angle of 45° with the V.P. Draw the projections of the cone when the apex is towards the observer. (Jan-2025-New) [NLJIET]	7	CO3	AN
Group-IV: CYLINDER				
14.	A cylinder, diameter of base 60 mm and height 70 mm, is having a point of its periphery of base in V.P. with axis of cylinder inclined to V.P. by 45° and parallel to H.P. or ground. Draw the	4	CO3	AN

	projections of the cylinder.(Jan-2025-New) [NLJIET]			
TOPIC:2 Section of Solids				
Group-I : PRISM				
15.	A square prism side of base 30 mm and its axis is 50 mm is kept on H.P. on its base such that two side of base are perpendicular to V.P. It is cut by the horizontal cutting plane which cuts the axis at its midpoint. Draw front view and sectional top view.(Mar-2022) [NLJIET]	3	CO3	AN
16.	A hexagonal prism, edge of base 25 mm and axis 65 mm long, rests with its base on H.P such that one of its rectangular faces is parallel to V.P. It is cut by a plane perpendicular to V.P and parallel to HP and 30mm above HP. Draw the development of the lateral surface of the truncated prism. (Jan-2025) [NLJIET]	4	CO3	AN
17.	One of the rectangular faces of the vertical square prism, 40 mm side and axis 65 mm long, make an angle of 30° to VP. It is cut by an AIP passing through the center of the axis in making an angle of 40° to HP. Draw front view, sectional top view and true shape of the section. (Jan-2020) [NLJIET]	7	CO3	AN
18.	A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. Draw the sectional top view.(Jun-2025-New) [NLJIET]	7	CO3	AN
Group –II : PYRAMID				
19.	A square pyramid, base 45 mm side and axis 70 mm long has its base in H.P. all edges of the base are equally inclined to V.P. It is cut by a section plane perpendicular to V.P. and inclined at 45° to the H.P. such that it bisects the axis. Draw its sectional top view and front view. (Aug-2022) [NLJIET]	4	CO3	AN
20.	Hexagonal pyramid of side of base 25 mm and height 55mm is resting on HP on its base. It is cut by a plane parallel to HP and perpendicular to VP and cutting the axis at 25 mm above the base. Draw its projection.(Jan-2024) [NLJIET]	4	CO3	AN
21.	A square pyramid, base 40 mm side and axis 60 mm long has its base on the HP and all edges of the base equally inclined to VP. It is cut by a section plane perpendicular to VP and inclined at 45° to the HP such that it bisects the axis. Draw its sectional top view and sectional left side view. (Jan-2020) [NLJIET]	7	CO3	AN
22.	A square base pyramid with 40 mm side and axis 65 mm long, has its base on the HP and all the edges of the base are equally inclined to the VP. It is cut by a section plane perpendicular to the VP and inclined at 45° to the HP. Further, it is bisecting the axis of the pyramid. Draw its sectional top view, sectional side view and true shape of the section.(Nov-2020) [NLJIET]	7	CO3	AN

23.	A pentagonal pyramid, side of base 25 mm and height 50 mm is resting on HP with one of the edge of base is perpendicular to VP. It is cut by auxiliary inclined plane inclined at 45° and cuts the axis at 20 mm above the base. Draw sectional top view, front view and true shape of section. (Jan-2024) [NLJIET]	7	CO3	AN
24.	A pentagonal pyramid, side of base 40 mm and height 80 mm, is resting on H.P. on its base with one of the edges of the base away from V.P. is parallel to VP. It is cut by an A.I.P. bisecting the axis, the distance of the section plane from the apex being 20 mm. Draw the elevation and sectional plan of the pyramid and draw the true shape of the section. Find the inclination of the section plane with the H.P. (Jan-2025-New) [NLJIET]	7	CO3	AN
Group-III : CONE				
25.	A cone, base diameter 30 mm and height 60 mm, is resting on its base. It is cut by AIP inclined at 45° to H.P. and passing through midpoint of the axis of cone. Draw front view, sectional top view and true shape of this sectional solid. (Mar-2023) [NLJIET]	7	CO3	AN
26.	A cone of 70 mm diameter of the base circle and 60 mm length of axis is resting on its base on the HP. It is cut by an AIP so that true shape of the section is an isosceles triangle with the vertex angle of 50° . Set the required cutting plane and find its inclination with the HP. Draw sectional top view, front view and project the true shape of the section. (Jun-2019) [NLJIET]	7	CO3	AN
27.	A cone having base diameter 80mm and height 90mm is resting with its base on H.P. It is cut by A.I.P. at 45° to the H.P. The cutting plane passes through the midpoint of the axis of the cone. Draw the elevation, the sectional plan and true shape of the section. [NLJIET]	7	CO3	AN
Group-IV: CYLINDER				
28.	A cylinder with 50 mm base diameter and 80 mm long axis, is lying on a generator on the H.P. with its axis parallel to the V.P. It is cut by an A.I.P. inclined at 30° to the H.P. and passes through a point on the axis 30 mm from one of its ends. Draw its front view AND sectional top view. (Aug-2023) [NLJIET]	4	CO3	AN
29.	A cylinder is resting on HP on its base. It is cut by AVP perpendicular to HP and inclined to VP by 45° and cutting it remaining 12 mm away from the axis. Draw the projections with section and draw also the true shape of the section. Take diameter of cylinder 55 mm and height 60 mm. (Jun-2019) [NLJIET]	7	CO3	AN

CHAPTER: 6

ORTHOGRAPHIC PROJECTIONS AND SECTIONAL ORTHOGRAPHIC PROJECTION

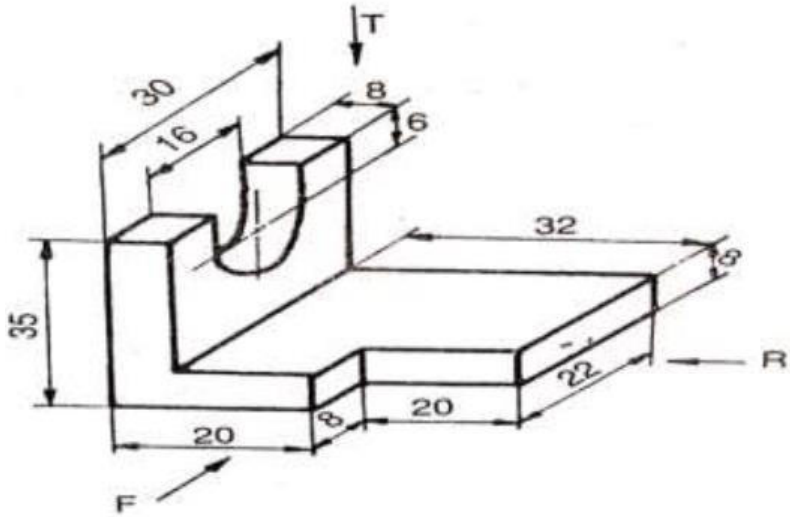
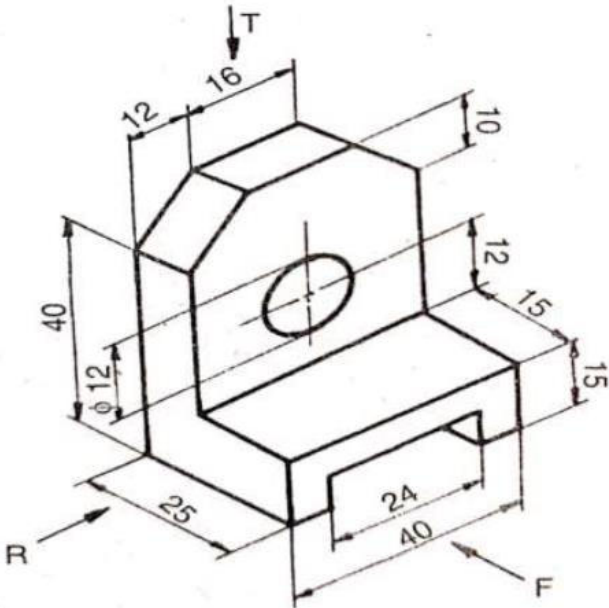
Fundamental of projection along with classification, principal planes for view from front, top and sides using first angle projection method and third angle projection method, full sectional view

SHORT QUESTIONS

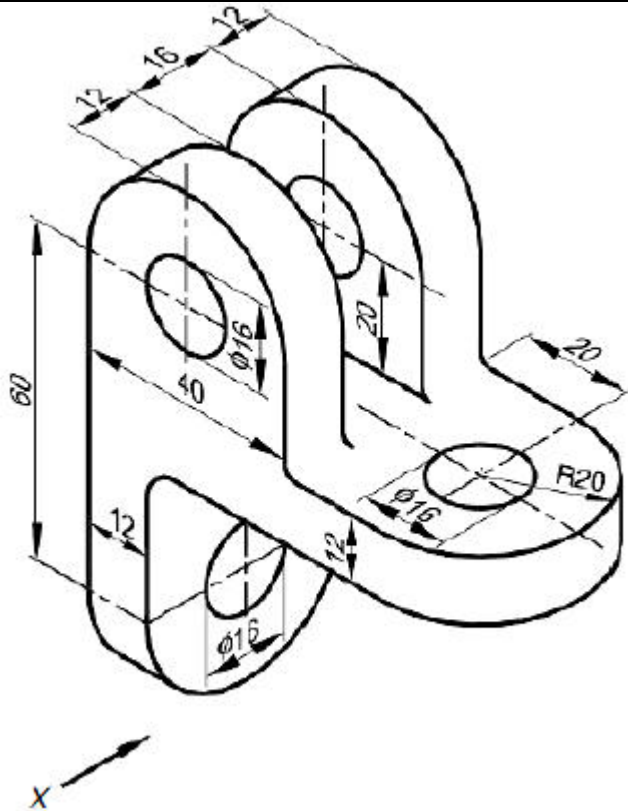
Sr. No.	QUESTIONS	Marks	CO	RBT Level
1.	Give symbols for methods of Projection.(Jan-2025-New) [NLJIET]	2	CO3	AN
2.	What is the difference (comparison) between a First Angle and Third Angle projection? (Jan-2020) [NLJIET]	3	CO3	AN
3.	Explain difference of any two projection methods. (Aug-2023) [NLJIET]	3	CO3	UN
4.	Write any three main differences between First angle and third angle projection system.(Nov-2020)[NLJIET]	3	CO3	UN
5.	Draw the conventional symbols for First Angle and Third Angle projection.(Jan-2020) [NLJIET]	3	CO3	UN
6.	Draw the symbols for first angle and third angle projection method. (Jul-2024) [NLJIET]	3	CO3	UN
7.	Write the comparison between first angle projection method and third angle projection method. (Aug-2022),(Jun-2025-New) [NLJIET]	4	CO3	UN
Sr. No.	QUESTIONS FOR DRAWING	Marks	CO	RBT Level
	TOPIC-1: Orthographic Projection			
1.	For the cuboid 50 x 30 x 20, draw all six orthographic views using 3 rd angle projection method. (Jan-2019) [NLJIET]	4	CO3	AN
2.	Using third angle projection method draw right hand side view for the object shown in Fig. (Jun-2019) [NLJIET]	4	CO3	AN

3.	Draw front view, top view and right-hand side view of the object given in Figure using first angle projection. (Nov-2020) [NLJIET]	7	CO3	AN
4.	Figure shows pictorial view of an object. Draw Front view looking in the direction F and also draw any one side view. (Mar-2022) [NLJIET]	7	CO3	AN

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5.	Figure shows pictorial view of an object. Draw Front view looking in the direction F and also draw any one side view. (Mar-2022) [NLJIET] 	7	CO3	AN
6.	Draw front view of an object shown in figure using first angle projection methods. (Mar-2023) [NLJIET] Draw top view and left hand side view of object shown in figure using first angle projection methods. (Mar-2023) [NLJIET]	4 7	CO3	AN

7.	Draw front view of an object shown in figure using first angle projection methods. (Mar-2023) [NLJIET] Draw top view and left hand side view of object shown in figure using first angle projection methods. (Mar-2023) [NLJIET]	4 7	CO3	AN
8.	Draw any three views using orthographic projections of fig shown below:(Aug-2023) [NLJIET]	7	CO3	AN

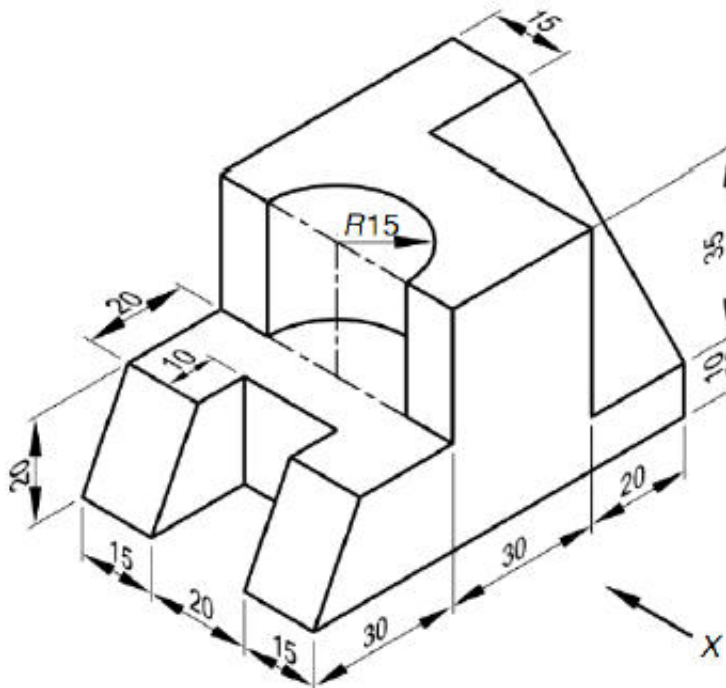


9. Draw the Front view of Fig. looking in the direction X and also draw any one side view. (Aug-2023) [NLJIET]

7

CO3

AN

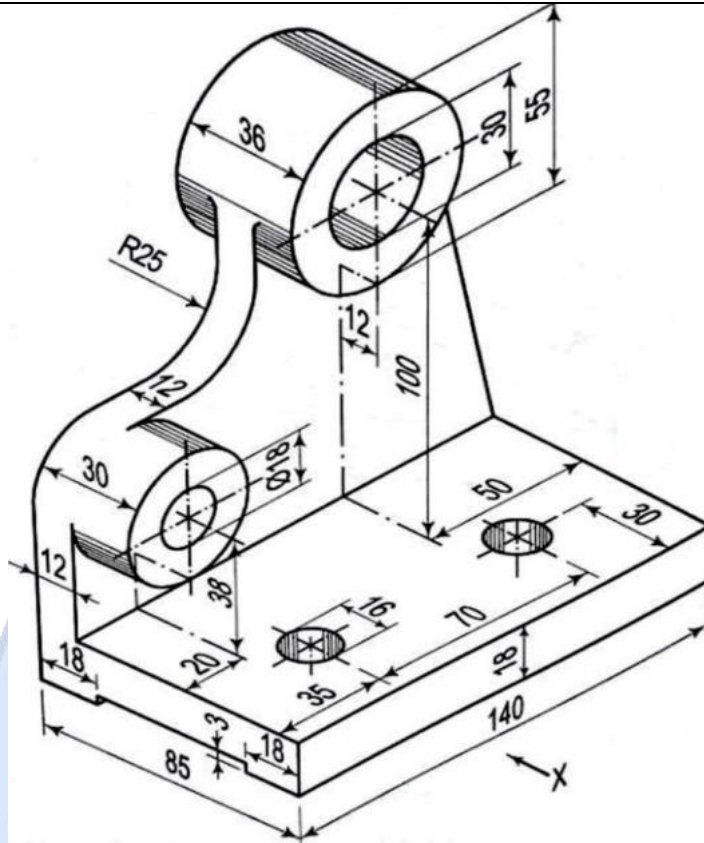


10. Draw the front view looking from X –direction and plan of a object shown in Figure using first angle projection method. (Jun-2025-New) [NLJIET]

7

CO3

AN

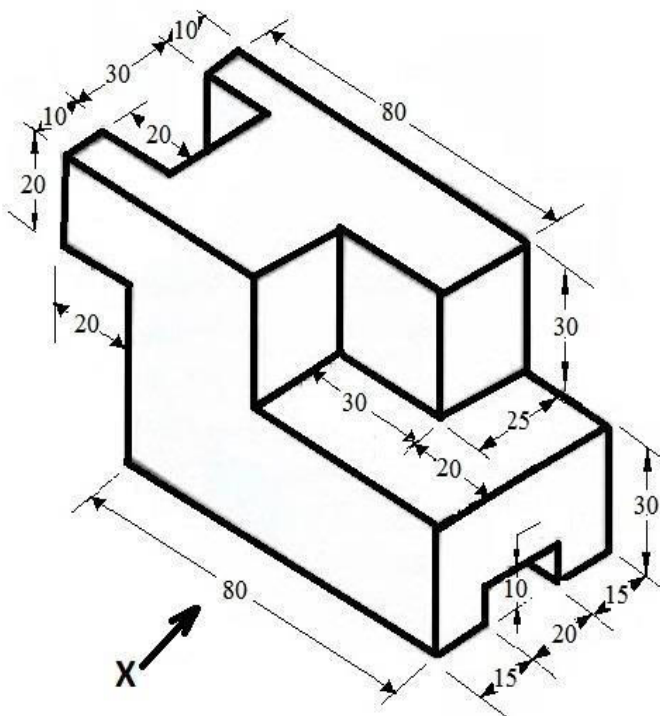


11. For the figure shown below, draw **plan**, elevation and right hand side view using third angle projection method. (Jan-2024) [NLJIET]

10

CO3

AN



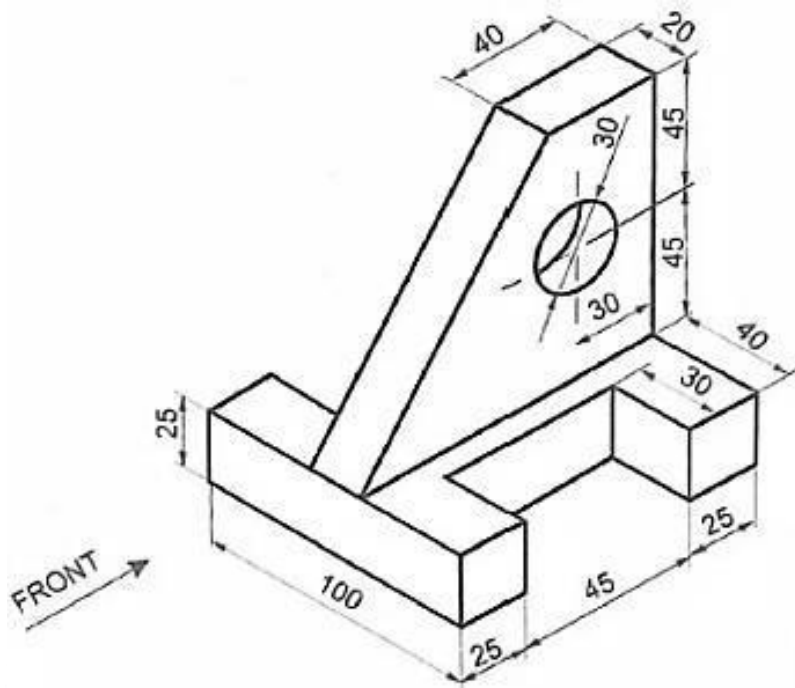
12. Draw orthographic projections of given figure (FV, TV, and R.H.S.V) as per first angle projection method.

10

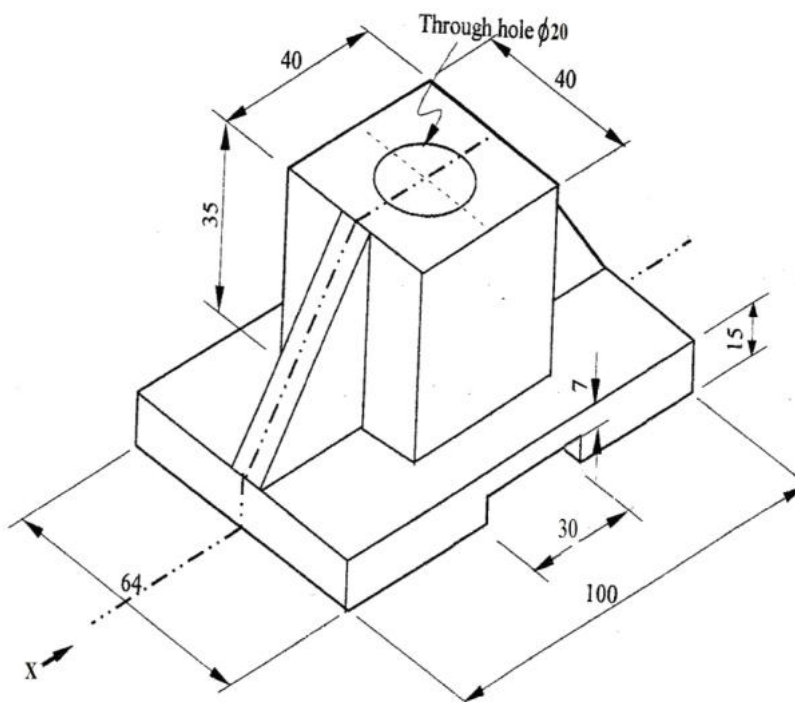
CO3

AN

(Jul-2024) [NLJIET]



13. Draw Front view, Top View and Right hand side view using 1st Angle projection method. (Jan-2025) [NLJIET]



TOPIC:2 Sectional Orthographic Projections

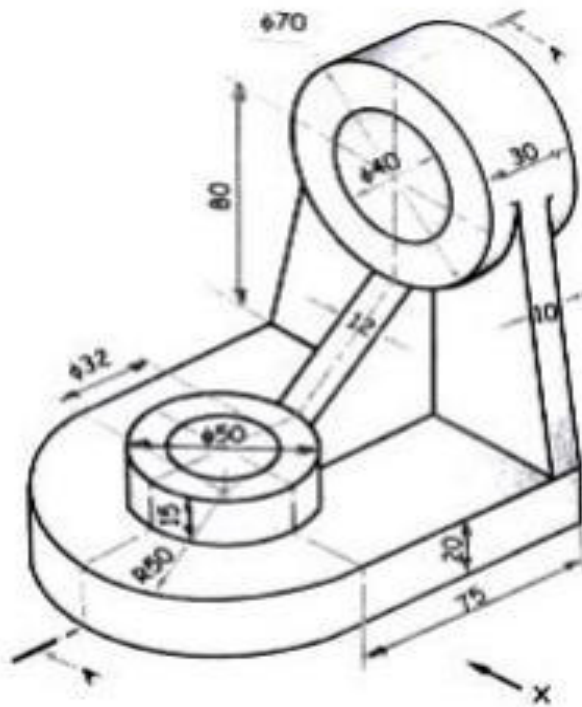
1. Figure shows the pictorial view of an object, draw the following views using first angle method of projection. Also give the

7

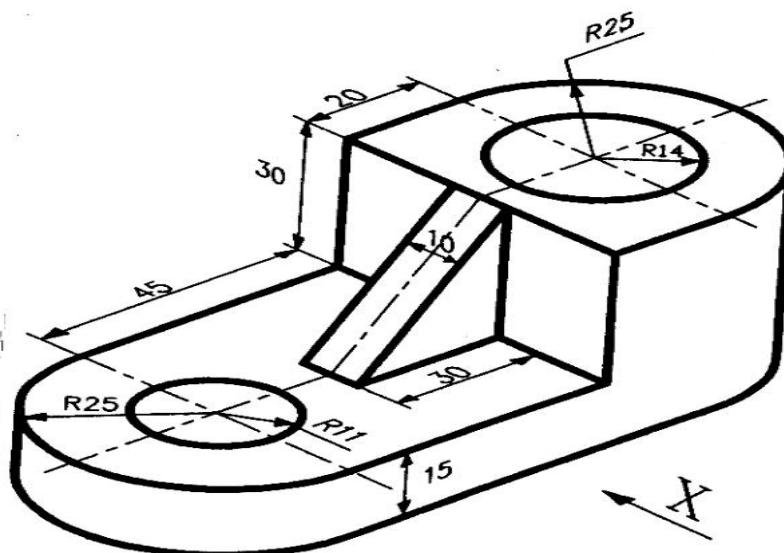
CO3

AN

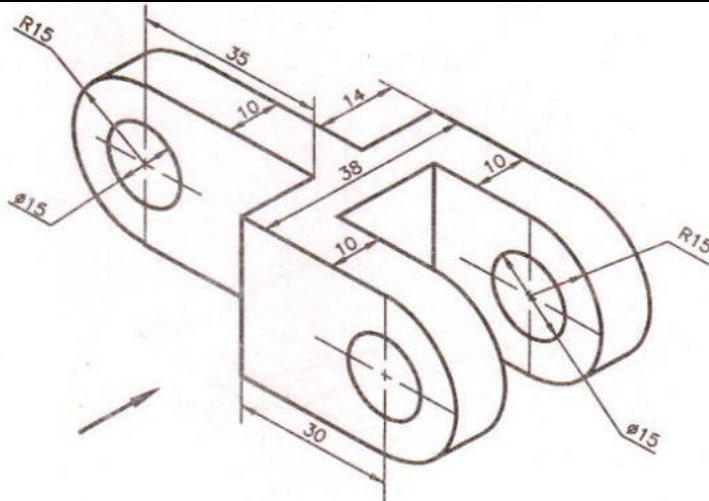
important dimensions. (i) Sectional front view, along section A-A (ii) Top View. (Nov-2020) [NLJIET]



2. Draw orthographic views of the figure by using first angle projection method: (i) Full sectional front view (ii) Top view (Aug-2022) [NLJIET]



3. Fig. 1 shows pictorial view of an object. Draw Sectional Elevation from -X and Plan. In the figure 1, the arrow shows "X" direction. (Jun-2025-New) [NLJIET]



4. Fig.1 shows pictorial view of an object. By using particular any one dimensioning method, draw following views by first angle projection method.(a) Sectional elevation at AA (b) Top view (c) Right hand side view (Jan-2025-New) [NLJIET]

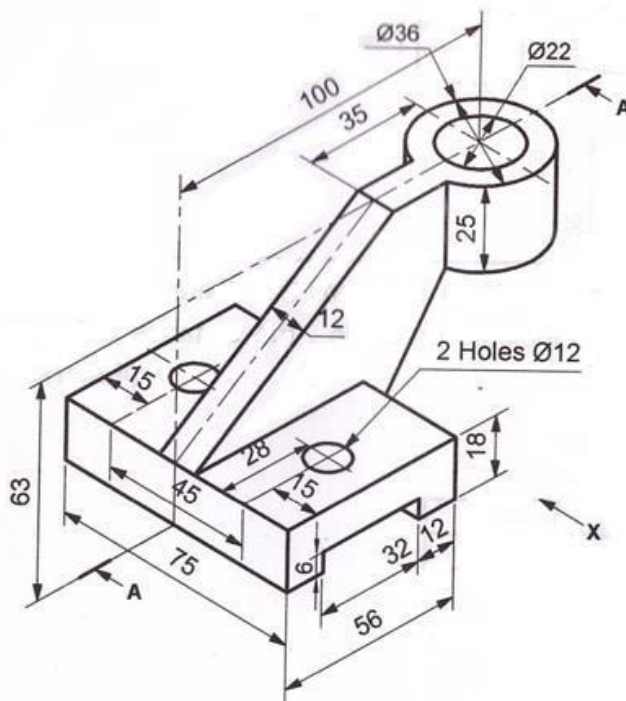


Fig.-1

5. Fig. shows pictorial view of an object. By using particular any one dimensioning method, draw following views by first angle projection method:(a) Sectional elevation at AA (b) Top view (c) Left hand side view (Jan-2025-New) [NLJIET]

10

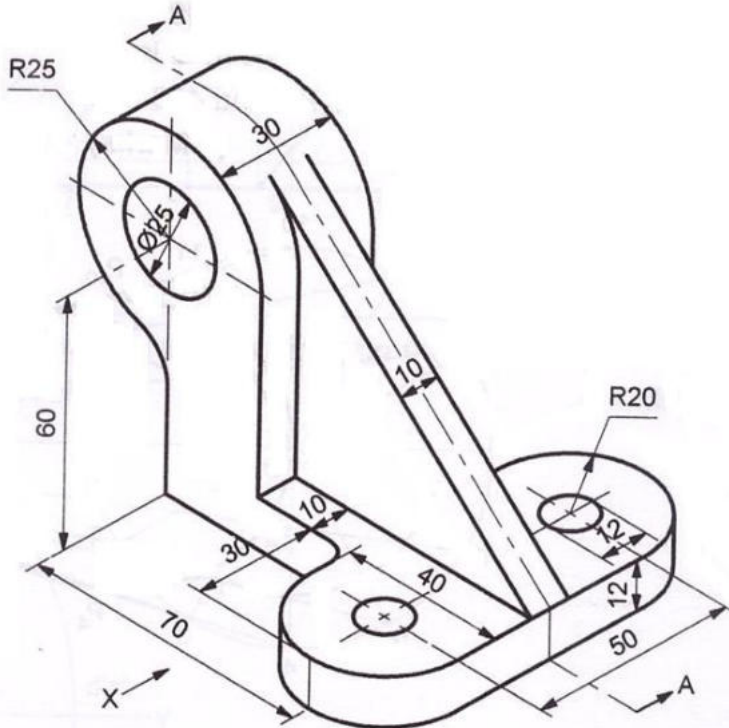
CO3

AN

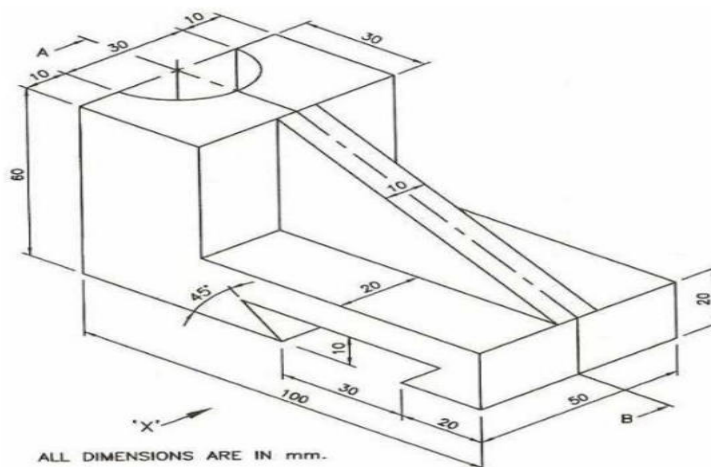
10

CO3

AN



6. Draw the following views for the Fig. using First angle projection method: (1) Sectional Elevation along A-B from X (2) Top view (3) RHSV. [NLJIET]



ALL DIMENSIONS ARE IN mm.

CHAPTER:7

COMPUTER AIDED DRAWING

Introduction to AutoCAD, Basic commands for 2D drawing like:

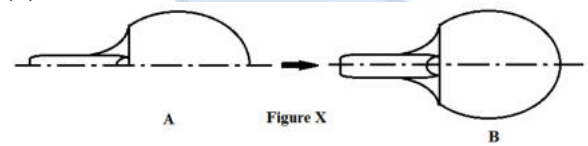
Line, Circle, Polyline, Rectangle, Hatch, Fillet, Chamfer, Trim, Extend, Offset, Dim style etc.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
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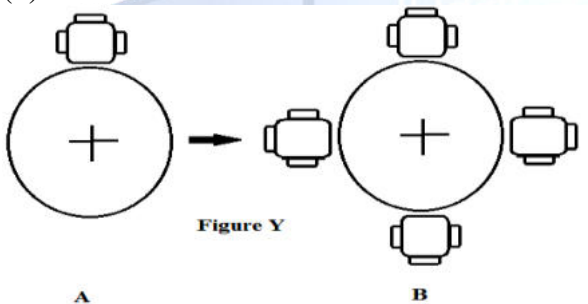
1.	Explain the following AutoCAD commands: hatch, circle and array.(Jun-2019) [NLJIET]	3	CO4	UN
2.	Why chamfer is done on work piece. Write the steps to create chamfer in AUTOCAD.(Jan-2019) [NLJIET]	3	CO4	RM
3.	Write (Comparison) difference between line, polyline and its uses in AUTOCAD. (Jan-2019), (Jan-2020) [NLJIET]	3	CO4	UN
4.	List the six Essential Commands of Modify Panel in AutoCAD. (Jan-2020) [NLJIET]	3	CO4	RM
5.	Explain following commands: (i) Line (ii) Poly Line (iii) Circle (Nov-2020) [NLJIET]	3	CO4	UN
6.	What are the (List) benefits of Computer Aided Drawing? (Nov-2020) [NLJIET]	3	CO4	RM
7.	State some of important commands used in auto CAD. (Mar-2022) [NLJIET]	3	CO4	RM
8.	Write the advantages of CAD. (Aug-2022) [NLJIET]	3	CO4	RM
9.	Write various commands used in AutoCAD and their usage. (Aug-2022) [NLJIET]	3	CO4	RM
10.	Explain any three AutoCAD commands in short (1) Spline (2) Rotate (3) Polygon (4) Multiline (Mar-2023) [NLJIET]	3	CO4	UN
11.	Explain any three AutoCAD commands in short. (1) Circle (2) Array (3) Mirror (4) Fillet Command. (Mar-2023) [NLJIET]	3	CO4	UN
12.	State the importance of AutoCAD in engineering field. (Aug-2023) [NLJIET]	3	CO4	UN
13.	Enlist any six CAD drawing preparatory commands with its symbols and function.(Aug-2023) [NLJIET]	3	CO4	UN
14.	Benefits of AUTOCAD over conventional drawing techniques. (Jan-2024) [NLJIET]	3	CO4	RM
15.	Which AUTOCAD command can be used to create Part B from Part A for the following figures?(Jan-2025) [NLJIET]	3	CO4	AN

(1)



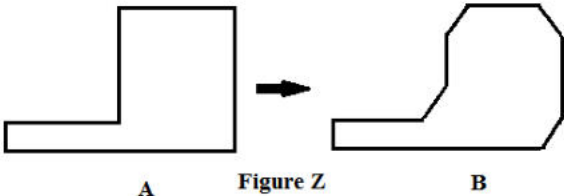
A Figure X B

(2)



A Figure Y B

(3)

	 A Figure Z B			
16.	Explain the following AutoCAD commands: mirror, trim, extend and fillet.(Jun-2019) [NLJIET]	4	CO4	UN
17.	List and explain different methods to draw rectangle in AUTOCAD. (Jan-2019) [NLJIET]	4	CO4	RM
18.	List and explain different methods to draw circle in AUTOCAD. (Jan-2019) [NLJIET]	4	CO4	RM
19.	List and explain different methods to draw arc in AUTOCAD. (Jan-2020) [NLJIET]	4	CO4	RM
20.	List and explain different methods to draw a polygon in AUTOCAD.(Nov-2020) [NLJIET]	4	CO4	RM
21.	List and explain different methods to draw Polygon in AUTOCAD. (Jan-2020) [NLJIET]	4	CO4	RM
22.	What are the various advantages of Auto CAD? (Mar-2022) [NLJIET]	4	CO4	RM
23.	What are advantages of AUTOCAD over conventional drawing? (Jan-2025) [NLJIET]	4	CO4	RM
24.	What are the various applications of Auto CAD? (Mar-2022) [NLJIET]	4	CO4	RM
25.	Identify the command, that used (1) to remove certain unwanted part (2) to show sectional view (3) to make a continuous line and 4) to make a circular shape at the edge of drawing. (Aug-2023) [NLJIET]	4	CO4	RM
26.	What are the advantages and disadvantages of using AUTOCAD?(Jan-2024) [NLJIET]	4	CO4	RM
27.	Explain the importance of file extension in the communication of drawing.(Aug-2023) [NLJIET]	4	CO4	UN
28.	Explain use of following AUTOCAD commands: (Jan-2024) [NLJIET] 1. TRIM 2. ARC 3. MIRROR 4. GRID	4	CO4	UN
29.	What is meant by ARRAY in AUTOCAD? What are types of ARRAYS? (Jul-2024) [NLJIET]	4	CO4	RM
30.	Write down different software used in design. (Jul-2024) [NLJIET]	4	CO4	RM
31.	Explain use of following AUTOCAD commands: [NLJIET] (1) MOVE (2) FILLET (3) OFFSET (4) CHAMFER	4	CO4	RM
32.	Discuss advantages of AUTOCAD over conventional drawing. [NLJIET]	4	CO4	RM

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